

Introduction

BACKGROUND & MOTIVATION

Inspecting tall structures, such as cell towers, wind turbines is dangerous, costly, and time-consuming, requiring rope access or scaffolding. Drone-based inspection offers a safer alternative: a UAV captures imagery around the structure and 3D Gaussian Splatting (3DGS) reconstructs a photorealistic 3D digital twin. Raw 3DGS reconstructions contain the entire scene (ground, sky, vegetation, surrounding buildings), not just the target. Defects (corrosion, cracks, missing bolts, deformed panels) must be found automatically within millions of Gaussian primitives. Manual review at that scale is impractical; inspectors need a tool that directs attention to the regions that matter.

PROBLEM STATEMENT

Given drone-captured imagery of a structure, how can we automatically isolate the target object from the surrounding scene in a 3DGS model, detect and localise structural anomalies, and present the results through an interactive 3D viewer that serves as a decision-support system for human inspectors?

Objectives & Benefits

OBJECTIVES

- Isolate the target structure from clutter using trajectory-guided spatial filtering.
- Detect anomalies automatically with RRX and CFAR thresholding.
- Cluster and characterise defects with location, bounding box, and severity metadata
- Build a decision-support viewer, a real-time 3DGS visualization tool where inspectors can navigate between anomaly clusters and take informed action

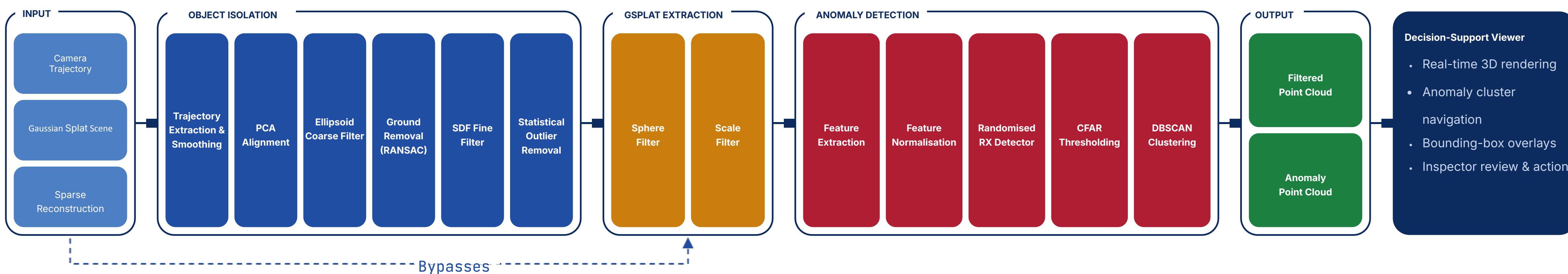
BENEFITS

Safety. Eliminates the need for human climbers; drones capture the data, the system does the analysis.

Fully automated. No manual annotation or labelled training data required; works on any structure type.

Decision-support, not replacement. The MVP highlights candidate defects for human experts to verify, combining AI speed with human judgement.

System Architecture



Methodology

STAGES 1 & 2 — STRUCTURE ISOLATION & GSPLAT EXTRACTION

Trajectory Smoothing

Raw COLMAP camera centres are smoothed by minimising a curvature-penalised least-squares energy.

Let P be the raw centres and D the second-difference operator:

$$\hat{p} = \operatorname{argmin}_x \|X - P\|^2 + \lambda \|DX\|^2$$

$$\Rightarrow (I + \lambda D^T D) \hat{p} = P \quad (\lambda = 10^7)$$

Solved independently per axis. Smoothed centres form the anchor set.

PCA Alignment & Ellipsoid Filter

Scene points are rotated into the trajectory's principal frame. An inflated ellipsoid ($\times 1.5$ radii, 95th-percentile extent) coarsely gates background:

$$C = (1/N)(X - \mu)^T(X - \mu) = V\Lambda V^T$$

$$\text{keep } p \text{ iff } p_x^2/a^2 + p_y^2/b^2 + p_z^2/c^2 \leq 1$$

RBF Signed Distance Field

RANSAC removes the ground plane (1 000 iter, inlier dist 0.1 m). A radial-basis SDF envelopes the structure; each anchor contributes three labelled samples (-1 inside, 0 surface, $+1$ outside):

$$K_{ij} = \exp(-\|x_i - x_j\|^2 / 2\sigma^2), \sigma = 20$$

$$(K + \epsilon I)w = s \Rightarrow \text{keep splat } q \text{ iff } d(q) < 0$$

SOR ($k = 50$, ratio ≤ 1.5) removes residual noise.

GSplat Extraction (Stage 2)

The sparse mask is transferred to the dense Gaussian model via two filters:

$$\text{Sphere: keep } q_j \text{ iff } \exists c : |q_j - c| \leq 0.5 \text{ m}$$

$$\text{Scale: keep } q_j \text{ iff } \max(\exp(\text{scale}_k)) < 0.5$$

STAGE 3 — AUTOMATED ANOMALY DETECTION

10-D Feature Representation

Each retained splat is encoded as a 10-dimensional vector and normalised with a robust scaler (median / IQR):

$$f = [x, y, z, f_{dc,0..2}, \text{opacity}, s_{0..2}] \in \mathbb{R}^{10}$$

Random Fourier Feature (RFF) Mapping

$D = 300$ random frequencies approximate the RBF kernel (σ estimated via median pairwise distances), lifting features to a 600-D space:

$$\omega_i \sim \mathcal{N}(0, \gamma^2 I_d), \gamma = 1/\sigma$$

$$z(x) = (1/\sqrt{D}) [\cos(\Omega x), \sin(\Omega x)] \in \mathbb{R}^{600}$$

Approximates $K(x, y) = \exp(-\|x - y\|^2 / 2\sigma^2)$ with $O(nD)$ complexity.

Randomised RX Anomaly Score

The Mahalanobis distance in RFF space measures how far each splat deviates from the background covariance structure:

$$G = (1/n) Z_c^T Z_c + \epsilon I, \epsilon = 10^{-6}$$

$$\delta(x^*) = z^{*T} G^{-1} z^*$$

CFAR Threshold & DBSCAN Clustering

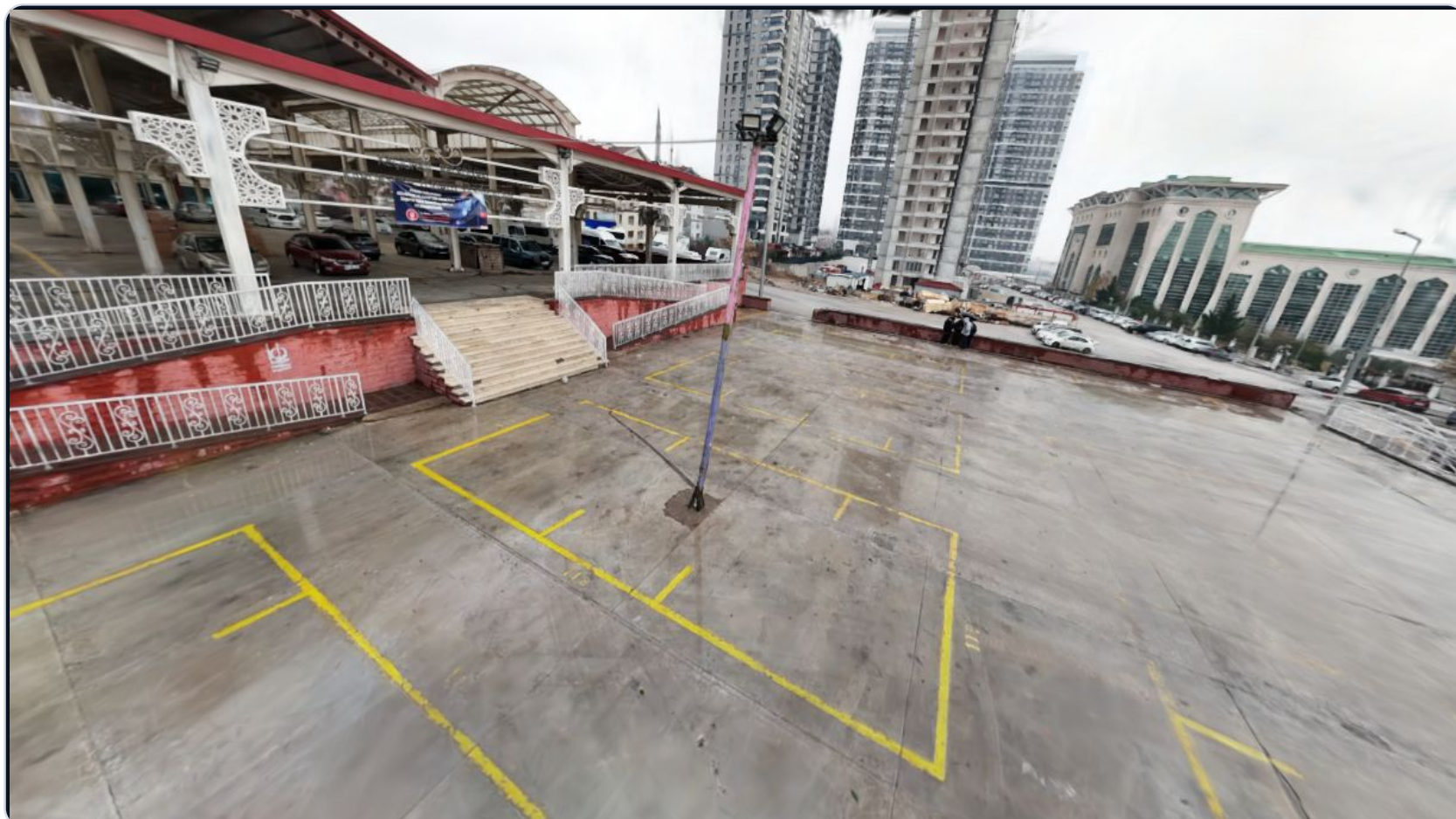
The best-fit distribution over $\{\Gamma, \log\text{-N}, \chi^2, \text{Weibull}, \text{Nakagami}, \dots\}$ is selected by KS test. The CFAR threshold controls the false-alarm rate exactly:

$$\tau = F^{-1}(1 - P_{fa}), P_{fa} = 0.01$$

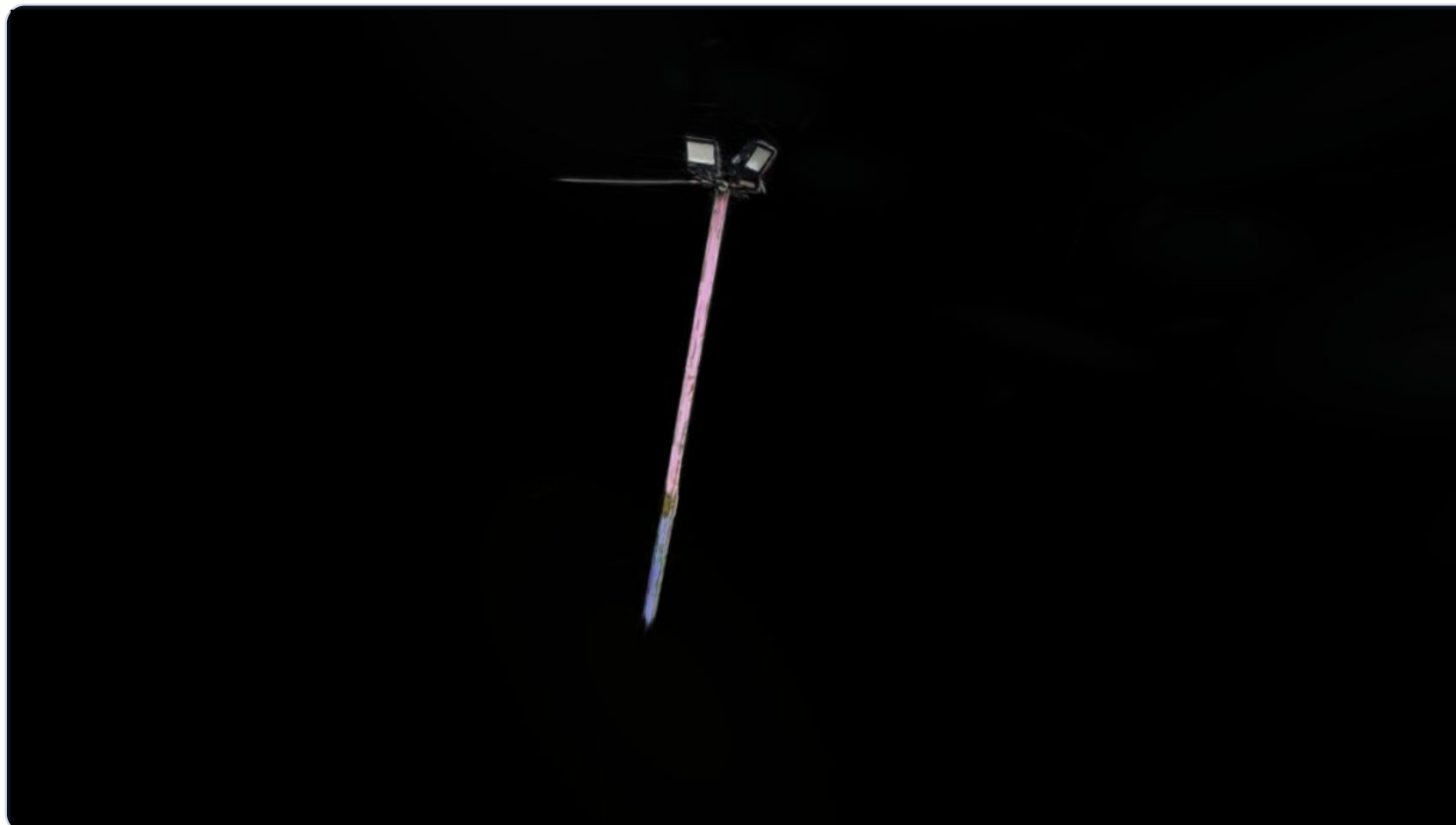
$$\text{anomaly} \Leftrightarrow \delta(x) > \tau$$

DBSCAN ($\epsilon=0.5$, min_samples=10) groups flagged splats into spatial clusters.

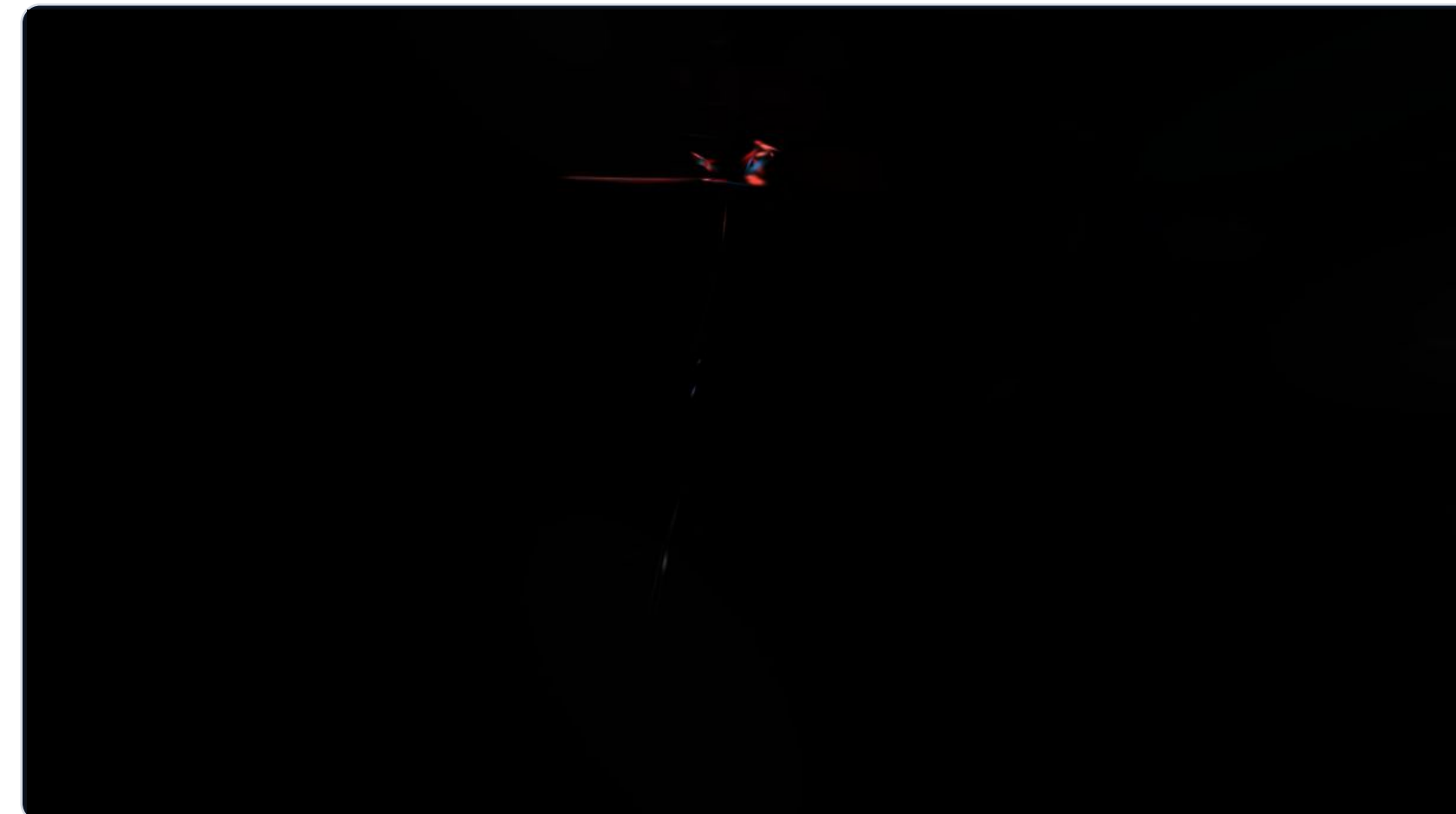
Results — Real-World Light-Pole Inspection



(a) Raw scene captured by drone — light pole and surrounding area as seen from UAV. Background clutter (ground markings, nearby buildings) dominates the raw 3DGS volume.



(b) Isolated 3DGS reconstruction — after Stage 1–2 trajectory-guided filtering. The lamp pole and fixtures are cleanly separated from background clutter without semantic supervision.



(c) Anomaly overlay — produced by Stage 3 (RRX + CFAR, $P_{fa} = 0.01$). The detected defect cluster is highlighted; DBSCAN groups the anomalous Gaussian primitives into spatially coherent regions for inspector review.

Conclusions

End-to-end pipeline isolates the structure, detects anomalies, and presents results for review.

Trajectory-guided filtering uses the drone's orbital path to remove background clutter.

Unsupervised anomaly detection uses Randomised RX with CFAR thresholding, requiring no labelled defect data.

Interactive 3D viewer helps inspectors review defect clusters and make informed decisions.

Future Work

Temporal change detection tracks defect progression across repeated 3DGS captures.

Defect classification labels anomaly types such as corrosion, cracks, missing parts, or vegetation.

Closed-loop autonomy guides drones to re-inspect flagged regions in close-up.

Cloud inspection platform enables browser-based review and automated report generation.

Outcomes

Project Repository

github.com/emredemirbas/
gaussian-splat-based-anomaly-detection



GITHUB

Related Publication

Camera-Trajectory-Driven Signed Distance
Fields for Automated Object Isolation
IEEE ICARA 2026
DOI:10.1109/ICARA69401.2026.11480350



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